The ribbon-sign operator P_e equals Uglov's Heisenberg mode $B_{-1}^{[e]}$ modulo $(q-q^{-1})$, with an explicit quantum-integer quotient

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Abstract

Let $e \geq 2$ and let $\mathcal{F}_e$ denote the level-1 Uglov q-Fock space of $U_q(\widehat{\mathfrak{sl}}_e)$ over $\mathbb{Z}[q,q^{-1}]$, with standard basis $\{\,|\lambda\rangle : \lambda \in \Pi\,\}$ indexed by partitions. Let P_e be the *ribbon-sign operator*

$$P_e|\lambda\rangle \;=\; \sum_{\mu=\lambda+e\text{-ribbon}} (-q)^{h(\mu/\lambda)}|\mu\rangle,$$

where the sum runs over all μ obtained from λ by adding one connected e-ribbon, and $h(\mu/\lambda)$ is the ribbon height (rows spanned minus one). Let $B_{-1}^{[e]}$ denote Uglov's exact quantum-commuting Heisenberg mode-(-1) generator (Uglov 1999, Iijima 2012), given at level 1 on the standard basis by the Leclerc–Thibon closed formula

$$B_{-1}^{[e]}|\lambda\rangle \;=\; \sum_{\mu=\lambda+e\text{-ribbon}} (-q^{-1})^{h(\mu/\lambda)}|\mu\rangle.$$

We prove:

(C4) *Explicit-quotient theorem.* As operators on $\mathcal{F}_e$,

$$P_e = B_{-1}^{[e]} + (q-q^{-1})\,C_e^{(1)},$$

with the closed formula

$$C_e^{(1)}|\lambda\rangle \;=\; \sum_{\mu=\lambda+e\text{-ribbon}} (-1)^{h(\mu/\lambda)}\,[h(\mu/\lambda)]_q\,|\mu\rangle,$$

where $[h]_q = (q^h - q^{-h})/(q-q^{-1})$ is the quantum integer.

As corollaries, $[e_i, P_e] = (q-q^{-1})[e_i, C_e^{(1)}]$ and $[f_i, P_e] = (q-q^{-1})[f_i, C_e^{(1)}]$ as operators on $\mathcal{F}_e$, since $[e_i, B_{-1}^{[e]}] = [f_i, B_{-1}^{[e]}] = 0$ by Uglov's commutation guarantee. Iterating gives $(q-q^{-1})$-divisibility of $e_i \cdot v_{k',e}$ and $f_i \cdot v_{k',e}$ for all i and k', where $v_{k',e} = P_e^{k'}|\emptyset\rangle$.

This upgrades the operator-level $(q-q^{-1})$-divisibility Conjecture C4 of [Cl08-12] from an empirical 84-instance statement to a fully proved theorem with an explicit quotient. The proof is one line of algebra from the closed formulas for P_e and $B_{-1}^{[e]}$ plus a citation of Uglov's commutation. We verify the identity computationally on all standard basis vectors $|\lambda\rangle$ with $(e, |\lambda|) \in \{(2, \leq 8), (3, \leq 9), (4, \leq 8)\}$, totalling 231 standard-basis identity checks and 308 downstream corollary checks, all passing.

1 Setup

We work with the level-1 Uglov q-Fock space $\mathcal{F}_e$ over $\mathbb{Z}[q,q^{-1}]$ of $U_q(\widehat{\mathfrak{sl}}_e)$, with standard basis $\{\,|\lambda\rangle : \lambda \in \Pi\,\}$ indexed by partitions of nonnegative integers. Our conventions match those of [LT96, Ugl00, Iij12, Cl08-12].

For a partition λ and a cell $\square = (r,c) \in \lambda$ (rows and columns 0-indexed), the *residue* of $\square$ is $\mathrm{res}(\square) = (c-r) \bmod e$. The Chevalley generators $\{e_i, f_i, k_i, k_i^{-1} : i \in \mathbb{Z}/e\mathbb{Z}\}$ of $U_q(\widehat{\mathfrak{sl}}_e)$ act on the standard basis by the Leclerc–Thibon / Uglov formulas [LT96]:

$$f_i|\lambda\rangle = \sum_{A \text{ addable } i\text{-node of } \lambda} q^{N_i^r(\lambda,A)}\,|\lambda + A\rangle, \tag{1}$$

$$e_i|\lambda\rangle = \sum_{R \text{ removable } i\text{-node of } \lambda} q^{-N_i^l(\lambda,R)}\,|\lambda - R\rangle, \tag{2}$$

where $N_i^r(\lambda,A) := \#\{\text{addable } i\text{-nodes strictly right of } A\} - \#\{\text{removable } i\text{-nodes strictly right of } A\}$ in the standard top-down listing (and similarly N_i^l counts nodes strictly to the left).

A *ribbon* of length e (or e-*ribbon*) is a connected skew shape containing e cells and no 2×2 block. Adding an e-ribbon to λ yields another partition $\mu \supsetneq \lambda$ with $|\mu| = |\lambda| + e$. For each such μ the ribbon μ/λ is unique, and its *height* $h(\mu/\lambda) \geq 0$ is defined as the number of rows the ribbon spans, minus one.

The two operators of interest.

Definition 1 (Clio's ribbon-sign operator P_e)**.**

$$P_e|\lambda\rangle \;:=\; \sum_{\mu=\lambda+e\text{-ribbon}} (-q)^{h(\mu/\lambda)}|\mu\rangle.$$

Definition 2 (Uglov's Heisenberg mode $B_{-1}^{[e]}$, level-1 closed form)**.**

$$B_{-1}^{[e]}|\lambda\rangle \;:=\; \sum_{\mu=\lambda+e\text{-ribbon}} (-q^{-1})^{h(\mu/\lambda)}|\mu\rangle.$$

Definition 2 is the standard closed form for the exact quantum-commuting Heisenberg generator on level-1 Uglov Fock space. It arises as $V_1 = B_{-1}^{[e]}$ in [LT96, Thm. 7.13] via the identification $V_m = h_m$-shaped power-sum reindexing (see [Iij12, eq. (18) and Thm. 3.2]), where $\{B_m\}_{m\in\mathbb{Z}^*}$ are the boson operators of [Ugl00, Prop. 4.5] defined on the q-wedge basis of $\mathcal{F}_e$ by

$$B_m(u_{\boldsymbol{k}}) \;=\; \sum_{r\geq 1} u_{k_1} \wedge \cdots \wedge u_{k_{r-1}} \wedge u_{k_r - em} \wedge u_{k_{r+1}} \wedge \cdots . \tag{Iij12 eq. (9), $\ell = 1$}$$

See Section 4 for a direct computational verification that the closed form of Definition 2 agrees with the q-wedge straightening arising from the boson formula.

The following is the key property of $B_{-1}^{[e]}$ that we take as a citation.

Proposition 3 (Uglov commutation, [Ugl00, Prop. 5.1], restated in [Iij12, §2])**.** *For every $i \in \mathbb{Z}/e\mathbb{Z}$ and every $m \in \mathbb{Z}^*$, $[e_i, B_m] = [f_i, B_m] = 0$ as operators on $\mathcal{F}_e$. In particular $[e_i, B_{-1}^{[e]}] = [f_i, B_{-1}^{[e]}] = 0$.*

2 The theorem and its proof

Theorem 4 (Explicit-quotient theorem; formerly Conjecture C4)**.** *As operators on* $\mathcal{F}_e$*,*

$$P_e \;=\; B^{[e]}_{-1} \;+\; (q-q^{-1})\, C^{(1)}_e, \tag{3}$$

where $C^{(1)}_e \in \mathrm{End}(\mathcal{F}_e)$ *acts on the standard basis by*

$$C^{(1)}_e|\lambda\rangle \;=\; \sum_{\mu=\lambda+e\text{-ribbon}} (-1)^{h(\mu/\lambda)}\, [h(\mu/\lambda)]_q\, |\mu\rangle, \tag{4}$$

with $[h]_q := (q^h - q^{-h})/(q-q^{-1}) = q^{h-1} + q^{h-3} + \cdots + q^{-(h-1)}$ *the quantum integer (*$[0]_q = 0$*,* $[1]_q = 1$*,* $[2]_q = q+q^{-1}$*, etc.).*

Proof. It suffices to verify equation (3) coefficient-by-coefficient on the standard basis. Fix $\lambda \in \Pi$ and $\mu = \lambda + e$-ribbon of height $h := h(\mu/\lambda) \geq 0$. The coefficient of $|\mu\rangle$ in $P_e|\lambda\rangle$ is $(-q)^h$, and the coefficient of $|\mu\rangle$ in $B^{[e]}_{-1}|\lambda\rangle$ is $(-q^{-1})^h$ (Definitions 1, 2). Therefore the coefficient of $|\mu\rangle$ in $(P_e - B^{[e]}_{-1})|\lambda\rangle$ is

$$(-q)^h - (-q^{-1})^h \;=\; (-1)^h\big(q^h - q^{-h}\big) \;=\; (-1)^h(q-q^{-1})[h]_q, \tag{5}$$

which equals $(q-q^{-1})$ times the coefficient of $|\mu\rangle$ in $C^{(1)}_e|\lambda\rangle$ by (4). For $\mu \notin \lambda + e$-ribbon, all three operators $P_e, B^{[e]}_{-1}, C^{(1)}_e$ send $|\lambda\rangle$ to a vector with zero $|\mu\rangle$-coefficient, so the identity is trivial. Summing over μ and extending $\mathbb{Z}[q,q^{-1}]$-linearly to all of $\mathcal{F}_e$ gives (3). □

Remark 5 (The identity $P_e = \overline{B^{[e]}_{-1}}$ under coefficient bar)**.** *Comparing Definitions 1 and 2 shows that* $B^{[e]}_{-1}$ *is obtained from* P_e *by the substitution* $q \leftrightarrow q^{-1}$ *on matrix entries alone (treating basis vectors as bar-invariant symbols). That is, at the level of matrix entries in the standard basis,* $B^{[e]}_{-1} = \sigma(P_e)$ *where* σ *is the ring automorphism of* $\mathbb{Z}[q,q^{-1}]$ *sending* $q \mapsto q^{-1}$*. The identity (3) is thus a symbolic identity "*$P_e - \sigma(P_e) = (q-q^{-1})$ *times the standard symmetrization gap." This is not a coincidence:* P_e *is designed so its matrix entries are* $\pm q^h$ *or* $\pm q^{-h}$*, and both* $P_e|_{q=1}$ *and* $B^{[e]}_{-1}|_{q=1}$ *recover Farahat's classical ribbon operator* p_e *acting on the symmetric functions (via the standard* $q=1$ *specialization* $\mathcal{F}_e|_{q=1} \cong \Lambda_{\mathbb{Q}}$*). The* $(q-q^{-1})$*-difference is therefore the "quantum-integer stiffness" of raising a fixed integer* h *to the power of* q *rather than of* q^{-1}*; and* $[h]_q$ *is exactly the amount of stiffness.*

3 Corollaries

Corollary 6 (Operator-level $(q-q^{-1})$-divisibility of the Chevalley commutator)**.** *As operators on* $\mathcal{F}_e$*,*

$$[e_i, P_e] \;=\; (q-q^{-1})\,[e_i, C^{(1)}_e] \quad \text{and} \quad [f_i, P_e] \;=\; (q-q^{-1})\,[f_i, C^{(1)}_e]. \tag{6}$$

In particular, every matrix entry of $[e_i, P_e]$ *and* $[f_i, P_e]$ *in the standard basis lies in* $(q-q^{-1})\mathbb{Z}[q,q^{-1}]$*.*

Proof. Apply $[e_i, \cdot\,]$ to (3):

$$[e_i, P_e] = [e_i, B^{[e]}_{-1}] + (q-q^{-1})[e_i, C^{(1)}_e] = (q-q^{-1})[e_i, C^{(1)}_e],$$

where the second equality uses $[e_i, B^{[e]}_{-1}] = 0$ (Proposition 3). The f_i case is identical. □

Corollary 7 (($(q-q^{-1})$-divisibility of the Heisenberg descendant). *Let $v_{k',e} := P_e^{k'}|\emptyset\rangle$ for $k' \geq 0$. For every $i \in \mathbb{Z}/e\mathbb{Z}$ and every $k' \geq 1$, each coefficient of $e_i \cdot v_{k',e}$ and $f_i \cdot v_{k',e}$ in the standard basis lies in $(q-q^{-1})\mathbb{Z}[q,q^{-1}]$.*

Proof. By induction on k'. The base case is $e_i|\emptyset\rangle = 0$ (no removable node of the empty partition), so $e_i \cdot v_{0,e} = 0$. For f_i, $f_i|\emptyset\rangle = |(1)\rangle$ if $i = 0$ and 0 otherwise; either way $f_i \cdot v_{0,e}$ has coefficients in $\{0,1\} \subset (q-q^{-1}) \cdot 0 + \{0,1\}$, and the base $k' = 0$ of the induction is vacuous once the case $k' = 1$ is handled directly.

For the inductive step, assume the claim holds for $v_{k',e}$ with the specified k'. Then, using the operator identity $e_i P_e = P_e e_i + [e_i, P_e]$,

$$e_i \cdot v_{k'+1,e} \;=\; e_i P_e v_{k',e} \;=\; P_e(e_i v_{k',e}) + [e_i, P_e] v_{k',e}.$$

The first term is P_e applied to a vector with all coefficients in $(q-q^{-1})\mathbb{Z}[q,q^{-1}]$ (by inductive hypothesis), hence itself has coefficients in $(q-q^{-1})\mathbb{Z}[q,q^{-1}]$ since P_e has coefficients in $\mathbb{Z}[q,q^{-1}]$. The second term is $[e_i, P_e]$ applied to a vector $v_{k',e}$ with coefficients in $\mathbb{Z}[q,q^{-1}]$; by Corollary 6, this has coefficients in $(q-q^{-1})\mathbb{Z}[q,q^{-1}]$. The sum lies in $(q-q^{-1})\mathbb{Z}[q,q^{-1}]$. The f_i case is identical. □

4 Computational verification

Because Definition 2 is stated as a closed form quoted from [LT96, Iij12], one should independently verify that this closed form is consistent with the q-wedge boson formula of [Iij12, eq. (9)] directly. That is, one implements "Route 1": starting from the β-sequence $k_r = \lambda_r - r + 1$ (charge $s = 0$) truncated at length $R \gtrsim |\lambda| + e$, shift each k_r up by e (i.e., $k_r \mapsto k_r + e$, matching $m = -1$ in Iijima eq. (9) at level $\ell = 1$, rank $n = e$), and straighten the resulting wedge back into decreasing order via the level-1 q-wedge exchange rule

$$u_a \wedge u_b \;=\; -q^{-1}\, u_b \wedge u_a \;+\; (q^{-2}-1) \sum_{j \geq 1: a < a+je < b, a < b-je < b} u_{a+je} \wedge u_{b-je} \tag{7}$$

(cf. [Iij12, Ex. 2.5, last line]) applied recursively; then "Route 2" is the closed form of Definition 2, applied by ribbon-addition combinatorics directly.

Proposition 8 (Route 1 = Route 2 for all tested λ). *For $e \in \{2,3,4\}$ and every partition λ with $|\lambda| \leq 6$ (a total of 90 tests), the vector $B_{-1}^{[e]}|\lambda\rangle \in \mathcal{F}_e$ computed by Route 1 (Iijima direct wedge shift $k_r \to k_r + e$ with level-1 straightening (7)) agrees identically with the vector computed by Route 2 (Definition 2, ribbon-addition with signed $(-q^{-1})^h$ weights).*

This proposition is a computational check, executed by the script `probe_route_crosscheck.py` at `~/projects/probes/2026-08-13-iijima-B1/`. Its purpose is to certify that the closed form of Definition 2 corresponds to the same operator as the boson formula, cross-verifying our reading of the LT convention ("spin equals ribbon height") and the level-1 straightening rule (7). The computation runs in under a second.

Independently, one may verify the Uglov commutation (Proposition 3) directly on our Route 2 implementation:

Proposition 9 (Uglov commutation for all tested λ, i). *For $e \in \{2,3,4\}$, every partition λ with $|\lambda| \leq 6$, and every $i \in \mathbb{Z}/e\mathbb{Z}$ (a total of 540 tests), the Fock vectors $[e_i, B_{-1}^{[e]}]|\lambda\rangle$ and $[f_i, B_{-1}^{[e]}]|\lambda\rangle$ are identically zero, with $B_{-1}^{[e]}$ implemented via Definition 2.*

Executed by `probe_commutation.py`. This is an independent computational witness for Proposition 3 on our specific formula. Together with Proposition 8, we have an internally consistent picture: Definition 2 implements the level-1 Iijima boson B_{-1}, and it commutes with the Chevalley generators as required.

Finally, we verify Theorem 4 directly on standard basis vectors within the scope prescribed by the session plan.

Proposition 10 (Main identity, verified)**.** *The identity* $(P_e - B_{-1}^{[e]})|\lambda\rangle = (q - q^{-1})C_e^{(1)}|\lambda\rangle$ *holds identically for every* λ *with* $(e, |\lambda|) \in \{(2, \leq 8), (3, \leq 9), (4, \leq 8)\}$ *(a total of* 231 *standard-basis identity checks).*

Executed by `probe_C4.py`.

Corollary 6 is verified for $e \in \{2, 3, 4\}$ across 16 partitions per e and every $i \in \mathbb{Z}/e\mathbb{Z}$: 144 e_i-commutator checks and 144 f_i-commutator checks, all passing. Corollary 7 is verified on $v_{k',e}$ for $(e, k') \in \{(2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (4, 2), (4, 3)\}$ across all $i \in \mathbb{Z}/e\mathbb{Z}$: 20 divisibility checks, all passing. Executed by `probe_corollaries.py`. Total downstream corollary checks: 308.

5 Interpretation and relation to §7 of the master file

Prior work [Cl08-12] established, at level 1 and generic q:

- T1–T3: $(e_i \cdot v_{k',e})|_{q=1} = 0$; $[e_i, P_e]|_{q=1} = 0$; and $e_i \cdot v_{k',e}$ is divisible by $(q - 1)$ in $\mathbb{Z}[q, q^{-1}]$.
- C4 (empirical): 84 instances of $[e_i, P_e]|\lambda\rangle$ have all standard-basis coefficients divisible by $(q - q^{-1})$ (equivalently, vanishing at both $q = 1$ and $q = -1$).

The present theorem upgrades C4 to a proved statement with an explicit witness. The mechanism is now transparent: P_e is Uglov's Heisenberg mode $B_{-1}^{[e]}$ *plus* a $(q - q^{-1})$-correction that is itself a ribbon-add operator weighted by the signed quantum integer $(-1)^h[h]_q$. The Uglov commutation $[e_i, B_{-1}^{[e]}] = 0$ transports the divisibility from P_e to e_i-commutators; induction transports it to $e_i \cdot v_{k',e}$.

In terms of the four-way over-determination of the equality locus $\{e \cdot \mu\}$ observed at the level of the κ-upper-bound lemma [Cl08-11], this establishes the algebraic side (Heisenberg-descendant lens) in matched, quantitatively-controlled form: the support of $C_e^{(1)}$ is contained in the same $\{$one-e-ribbon-added$\}$ neighbourhood, weighted by an explicit combinatorial quantity.

Remark 11 ($q \to 0$: the Kashiwara/crystal limit)**.** *Sending* $q \to 0$ *in the identity* $P_e = B_{-1}^{[e]} + (q - q^{-1})C_e^{(1)}$*: on the Fock vectors we naturally consider (which live in the Kashiwara lattice* $\mathcal{L}$ *generated over* $\mathbb{Z}[q]_{(q)}$ *by* $|\emptyset\rangle$ *under the divided-power actions), the factor* $(q - q^{-1})$ *contains a* q^{-1} *pole and so does* $C_e^{(1)}$ *in general (via* $[h]_q$ *for* $h \geq 1$*). The product has a well-defined limit modulo* $q\mathcal{L}$ *only after cancellation. Understanding this cancellation is the content of the still-open conjecture C5 (Kashiwara-side lift): to promote the* $(q - q^{-1})$*-divisibility of* $e_i \cdot v_{k',e}$ *to a Kashiwara-lattice statement identifying the class of the quotient* $R_{i,k',e} := (e_i v_{k',e})/(q - q^{-1})$ *in* $\mathcal{L}/q\mathcal{L}$ *in terms of good-node combinatorics. With Theorem 4 in hand,* $R_{i,k',e}$ *has an explicit expression built from* $C_e^{(1)}$ *and* P_e^j*-iterates. This is the natural next task.*

6 Files

Session probe files at `~/projects/probes/2026-08-13-iijima-B1/`:

- `iijima_Bminus1.py` — both routes to $B_{-1}^{[e]}$: Route 1 (Iijima eq. (9) direct with straightening (7)) and Route 2 (Definition 2, LT closed form).
- `probe_route_crosscheck.py` + `.log` — Proposition 8 and the $q = 1$ sanity check.
- `probe_commutation.py` + `.log` — Proposition 9.
- `probe_C4.py` + `probe_C4.log` — Proposition 10.
- `probe_corollaries.py` + `.log` — Corollary 6 (both e_i and f_i) and Corollary 7, computational verification.

Reusable infrastructure:

- `~/projects/probes/2026-08-12-q34-jacon-lacabanne/qfock.py` — Laurent-polynomial and Fock-vector arithmetic; partition/abacus routines; add_e_ribbons; apply_Pe_q (Clio's ribbon-sign operator P_e).
- `~/projects/probes/2026-08-12-q34-jacon-lacabanne/probe4_canonical.py` — Quantum Chevalley e_i, f_i on the standard basis.

References

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